

Algorithm Approach & Solution

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Subject: Code Documentation (LC1162 / Multi-source BFS)

Strategy & Notes

- **Core Approach:** Using the same approach as the 01 matrix question (LC542). First, build the distance matrix of each cell using multi-source BFS (or 2D DP), and then traverse again to find the maximum.
- **Optimization:** If we do this using multi-source BFS, `ans++` is incremented after each level. We just need to reach the last level (i.e., an empty queue) and return `ans` at that point.
- **Multi-Source BFS Logic:** "ek saath saare land expand krna he , hence multisource" (Expand all land cells simultaneously).
- **State Tracking:** To mark cells as visited, we effectively make them land (`grid[nr][nc]=1`) as they are added to the queue.

Source Code implementation

```
class Solution {
public:
    int maxDistance(vector<vector<int>>& grid) {

        class="comment">//im using the same approach as 01 matrix question
        LC542
        class="comment">//we can first build the distance matrix of each
        cell using multisource bfs or 2d dp just like 01 matrix
        class="comment">//and then traverse again to find maximum

        class="comment">//however if i do using multisource bfs , then
        after each level ans++
        class="comment">//we just need to reach the last level (i.e empty
        queue) and return ans at that point

        int n=grid.size();
        int m=grid[0].size();
```

```

int ans=0;

vector<pair<int,int>> dir={{1,0},{-1,0},{0,-1},{0,1}};
queue<pair<int,int>> q;
int lund=0;
for(int i=0;i<n;i++){
    for(int j=0;j<m;j++){
        if(grid[i][j]==1){
            q.push({i,j});
            lund++;
        }
    }
}

if(lund==0) return -1;
if(lund==n*m) return -1;

while(!q.empty()){
    int currsz=q.size(); class="comment">//ek saath saare land
expand krna he , hence multisource
    while(currsz--){
        int r=q.front().first;
        int c=q.front().second;
        q.pop();
        for(auto it:dir){
            int nr=r+it.first;
            int nc=c+it.second;
            if(nr>=0 && nc>=0 && nr<n && nc<m){
                if(grid[nr][nc]==0){
                    grid[nr][nc]=1; class="comment">//to mark them
visited we making them land
                    q.push({nr,nc});
                }
            }
        }
    }

    if(!q.empty()){
        ans++;
    }
}

return ans;

```

```
}  
};
```

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